Dimitris Volteras, PhD

Computational Biologist | Machine Learning Scientist

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PROFILE

Computational biologist and machine learning scientist developing mathematically principled models of complex biological systems. My research combines mechanistic and probabilistic modelling with modern machine learning to study cellular dynamics, gene regulation and lineage relationships from high-dimensional single-cell and perturbational data. Experienced in developing generative, dynamical and Bayesian modelling frameworks from first principles, with strong foundations in stochastic processes, causal inference and representation learning. I work closely with experimental scientists to translate biological questions into computational models and hypothesis-driven research programmes.

CORE EXPERTISE

Machine Learning & Mathematical Modelling: Generative modelling | Representation and self-supervised learning | Variational methods | Dynamical models and neural ODEs/SDEs | Bayesian inference | Causal and mechanistic modelling | Stochastic processes | Probabilistic modelling

Computational & Single-Cell Biology: Single-cell transcriptomics (scRNA-seq) | Perturbational and interventional data (Perturb-seq, signalling screens) | Developmental biology | Lineage tracing | Metabolic RNA labelling | Gene regulatory networks | Cellular dynamics and trajectory inference

Programming & Computing: Python (PyTorch, NumPy, SciPy, Scikit-learn, Scanpy, scVI) | Julia | R | Mathematica | Distributed and multi-GPU training | Large-scale simulation | SLURM/HPC | Linux/Bash | Git/GitHub

RESEARCH EXPERIENCE

Postdoctoral Research Fellow | Francis Crick Institute, London

Developmental Dynamics Laboratory - Dr James Briscoe | Mar 2025 - Present

- Developed a mechanistically informed generative model of cell-state dynamics, combining an interpretable latent-variable model with dynamical constraints to infer regulatory structure from large-scale, multi-condition single-cell signalling perturbation data.
- Developed a probabilistic framework for inferring hierarchical lineage relationships from single-cell genomic barcoding data, contributing to a multidisciplinary study of neural tube developmental dynamics (Nature, accepted).
- Led development of a stochastic modelling framework to infer gene-regulatory mechanisms from time-resolved single-cell transcriptomic data integrating metabolic RNA labelling and developmental-stage information.
- Developing a self-supervised framework for modelling high-dimensional interventional single-cell data, inspired by joint-embedding predictive architectures.
- Built and optimised scalable ML and numerical-simulation pipelines on HPC infrastructure, including distributed multi-GPU model training.

Postdoctoral Research Associate | Imperial College London

Computational Molecular Systems Biology Group - Dr Vahid Shahrezaei | Sep 2024 - Feb 2025

- Developed a stochastic agent-based modelling framework to investigate the spatiotemporal dynamics of tumour progression.
- Contributed computational modelling and quantitative analysis to a multidisciplinary study identifying axonal injury as a targetable driver of glioblastoma progression (Nature, 2025).

RESEARCH EXPERIENCE CONTINUED

PhD Researcher in Computational Biology | Imperial College London

Computational Molecular Systems Biology & Single-Cell Dynamics Groups - Dr Vahid Shahrezaei & Dr Philipp Thomas | Oct 2020 - Sep 2024

- Developed stochastic and probabilistic methods to infer gene regulatory mechanisms from high-dimensional single-cell transcriptomic data.
- Developed a Bayesian inference framework for genome-wide transcriptional dynamics from time-resolved single-cell RNA sequencing, published in Cell Systems.
- Investigated how cell-to-cell variability and dynamical correlations can reveal modes of gene regulation from single-cell measurements.

EDUCATION

PhD in Applied Mathematics & Computational Biology | Imperial College London

Thesis: Data-driven stochastic modelling of gene regulation; Roth PhD Scholarship, Department of Mathematics | 2020 - 2024

MSc in Applied Mathematics | Imperial College London

Thesis: Multiscale approximations of stochastic reaction networks; focus: mathematical biology, machine learning, stochastic differential equations and Markov processes |

BSc in Mathematics | National and Kapodistrian University of Athens

Focus: probability theory, stochastic analysis, Bayesian inference, ODEs and PDEs, and linear algebra |

SELECTED PUBLICATIONS

- Volteras, D., Thomas, P., & Shahrezaei, V. (2024). Global transcription regulation revealed from dynamical correlations in time-resolved single-cell RNA sequencing. Cell Systems. DOI: 10.1016/j.cels.2024.07.002
- Boezio, G. L. M., Depotter, J. R. L., Frith, T. J. R., Radley, A., Volteras, D., Strohbuecker, S., et al. (2026). Hierarchical lineage architecture of human and avian spinal cord revealed by single-cell genomic barcoding. Nature. Accepted.
- Clements, M., Tang, W., Baronik, Z. F., Ragdale, S., Oria, R., Volteras, D., White, I. J., et al. (2025). Axonal injury is a targetable driver of glioblastoma progression. Nature. DOI: 10.1038/s41 586-025-09 411-2

SELECTED MANUSCRIPT IN PREPARATION

Volteras, D., Thomas, P., & Shahrezaei, V. Inferring causal gene regulatory relationships from time-resolved single-cell transcriptomics with mechanistic machine learning. Manuscript in preparation.

SELECTED AWARDS

- Best Project Award, AI Hackathon co-organised by Google DeepMind and the Francis Crick Institute (2026)
- Roth PhD Scholarship, Department of Mathematics, Imperial College London (2020)